UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis Survey
Homework 3: Python practice
Due Thursday Aug 29, 6pm.

Goals: Become sufficiently familiar with Python to:

1. Start Python
2. Allocate, manipulate, and get information about arrays,
3. Print the value of any cell in an array,
4. Change values in an array according to given mathematical or positional criteria (e.g., zero every 5th pixel, set all values greater than 100 to -2000, set a given rectangular sub-array to a given value, insert one array into another),
5. View images and manipulate image colormaps interactively from Python,
6. Make line plots and figures with images, write them in standard formats,
7. Use available resources to find functions, packages, and any instructions you need.

Readings:

1. Do the Matplotlib pyplot tutorial. See https://matplotlib.org/tutorials/introductory/pyplot.html.
2. Do sections 1 and 2 of the Using Python for Interactive Data Analysis tutorial. See Files/demos_lectures/week_2/pydatatut/. The exercises are for your edification (don't hand them in).
3. https://github.com/rougier/numpy-100 . The exercises are for your edification (don't hand them in).
4. See Files/ demos_lectures/week_2/ds9.py for displaying and manipulating images with ds9.

Do not use loops for any problem on this assignment.

Remember to open your `0-ast4762-<yourname>.log` and start logging what you are doing. Also, remember that '`<yourname>`' should be replaced by the username you have chosen.

NOTE: The problems here can be done with much less than the full reading assignment above. Be sure you have mastered the material needed to do the problems, and then get as far as you can in the reading assignment and its exercises. You will be using all of that material, and more, in the rest of the course. This is the only time budgeted for refreshing/learning Python.

1. (**5 points**) Use your terminal to make a new folder `hw3_<yourname>` under your `ast4762/homework` folder. Explain in your log how you did it. For this homework your main homework file is a Python file named `hw3_<yourname>.py` (if you prefer to work on a notebook you will start with a `hw3_<yourname>.ipynb` but remember to save it as a .py before submission; see end of homework too). Save it in your homework (hw3) folder. Remember to commit your files and push to GitHub as often as needed (also, great backup!).

Your name, assignment number, and the date should appear as comments at the top of the file. At the start of every problem print the problem number using the appropriate Python command. Any remarks or written answers you may make should be printed out using the appropriate Python command. If you need to comment something in the code (for coding clarity) do so with a normal comment (i.e., `# this is a comment`). Remember that using comments is good coding practice, and it will be taken into account during the grading process.

2. (**10 points**) Write the necessary Python commands that (remember: no loops!):

a) Create an array of integers `x` from 0 to 1000 (up to and including 1000). How many elements do you need?
   Print the datatype of the array and the array's minimum and maximum.
   Remember to answer all questions raised in the assignment, such as "how many elements do you need?".
b) Re-scale `x` to contain values from 0 to 2π. ***Don't make a new array***, re-scale the array of question 2a.
   Print the minimum and maximum values of the new `x` array.
c) Make array `y` whose values are the sine values of `x`.
d) Print the value of element 234 of `y` [Note the difference between the 234th element and element 234 in Python!].

3. (**10 points total**) Write the necessary Python commands that:

a) (**5 points**) Plot `y` vs. `x` from problem 2c. Make the plot publication-ready using reasonable axis labels etc.
b) (**5 points**) Save your plot as a PNG using the appropriate Python commands (no screenshots or window dumps from outside Python).

4. (**10 points total**)

a) (**5 points**) Make a "ramp" array `r` with 101 evenly spaced elements going from -1 to +1. "Clip", or mask, the array so that any value greater than 0.5 is set to 0.5 and any value less than -0.5 is set to -0.5.

   There are different ways that you can do this with numpy, but remember, no loops!

b) (**5 points**) In the same plot, plot the original and clipped arrays. Save a screen shot that includes both the Python code you used to make the plot and the plot. Use an appropriate name for your screenshot.

5. (**10 points**) Give the URLs of two web sites outside of UCF that provide free astronomical software that is written in Python. Write a paragraph about each package in your own words. Put the two paragraphs as an extended string (between sets of triple single-quotes) in your main homework file.

6. (**10 points**) Prepare and submit your homework. If you worked on a Jupyter notebook, make sure to save it as a.py file. Don't forget to commit and push to GitHub. Explain what you did to do that in your log. Make a screenshot that shows your logging history, showing that you committed the file and add it to your `hw3_<yourname>` folder (remember to use an appropriate name for the screenshot!). Then:

- use a terminal to navigate to `ast4762/`. Explain how you did this in your log.
- Use the appropriate command to copy the `hw3_yourname` folder from your `homework` into your `handin` folder. Explain how you did this in your log.
- Go into your `handin/hw3_yourname` and run your .py from the terminal. Does it run without issues?
- Write what command (no drag and drop!) you will use to copy your log into your `handin/hw3_yourname`, make a tar.gz of the `handin/hw3_yourname` folder and how you will submit that to Webcourses.
- Use the appropriate commands to copy (from the terminal) your log into your `handin/hw3_yourname` folder, and make tar.gz of the folder.
- Turn the file in on WebCourses.